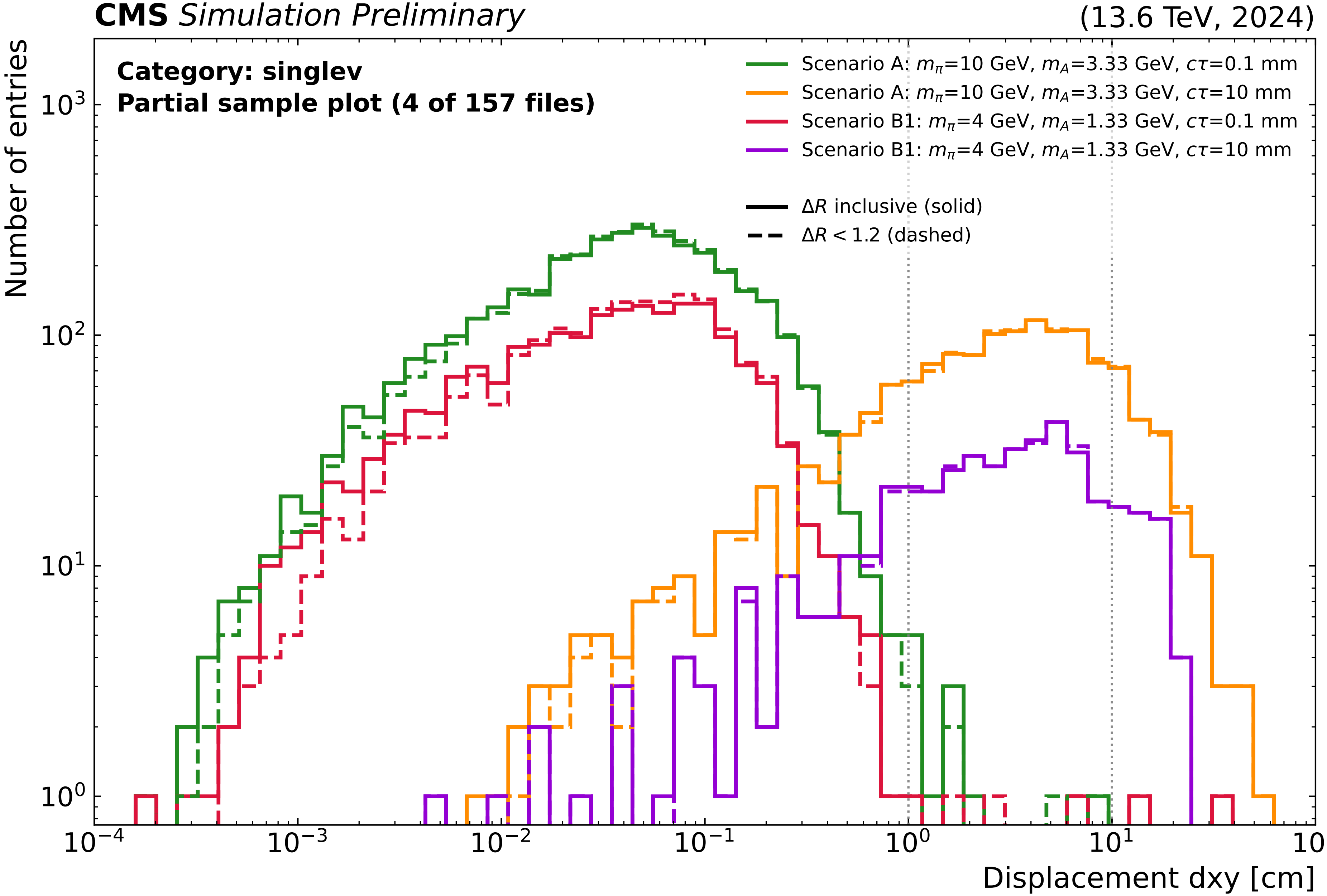
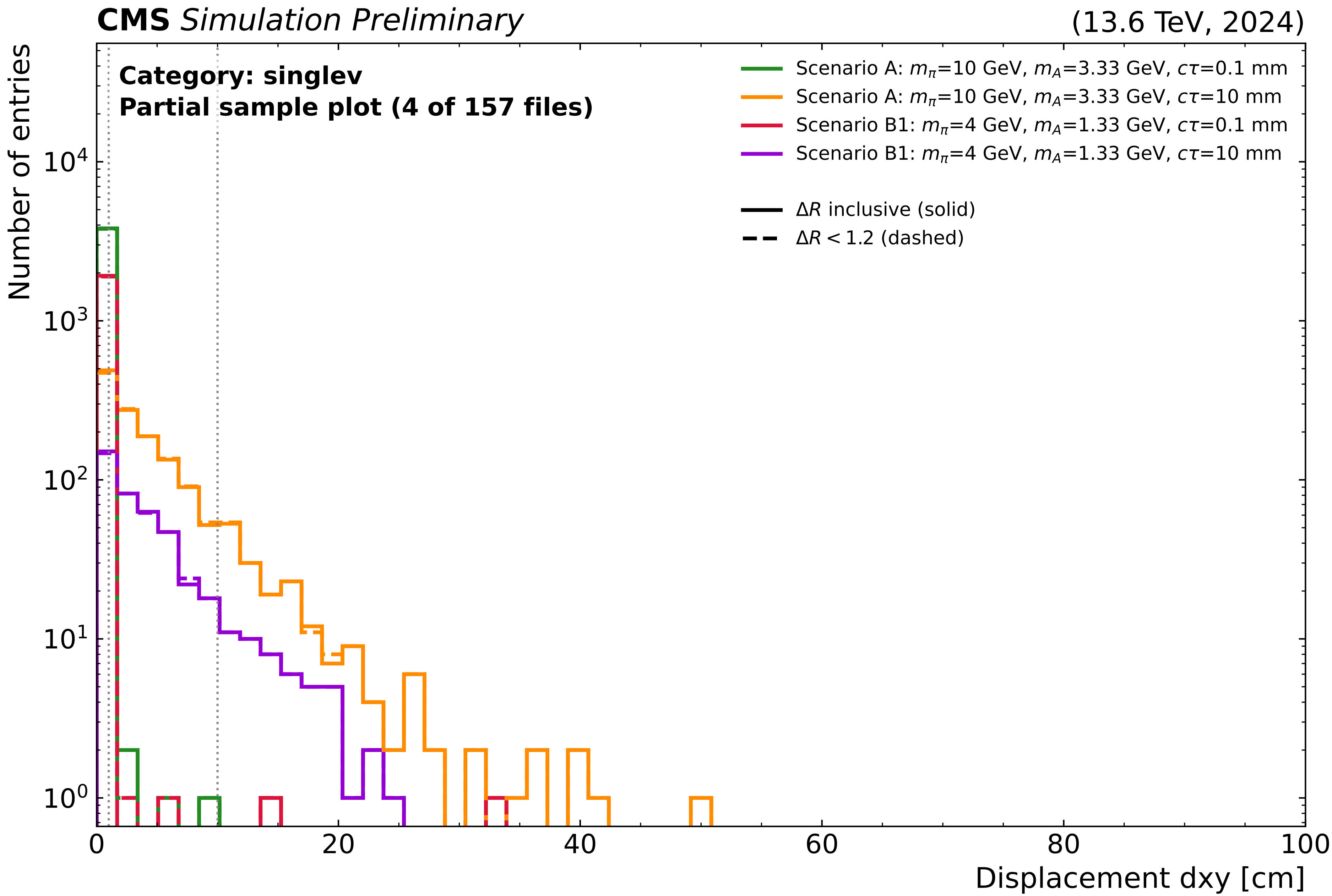
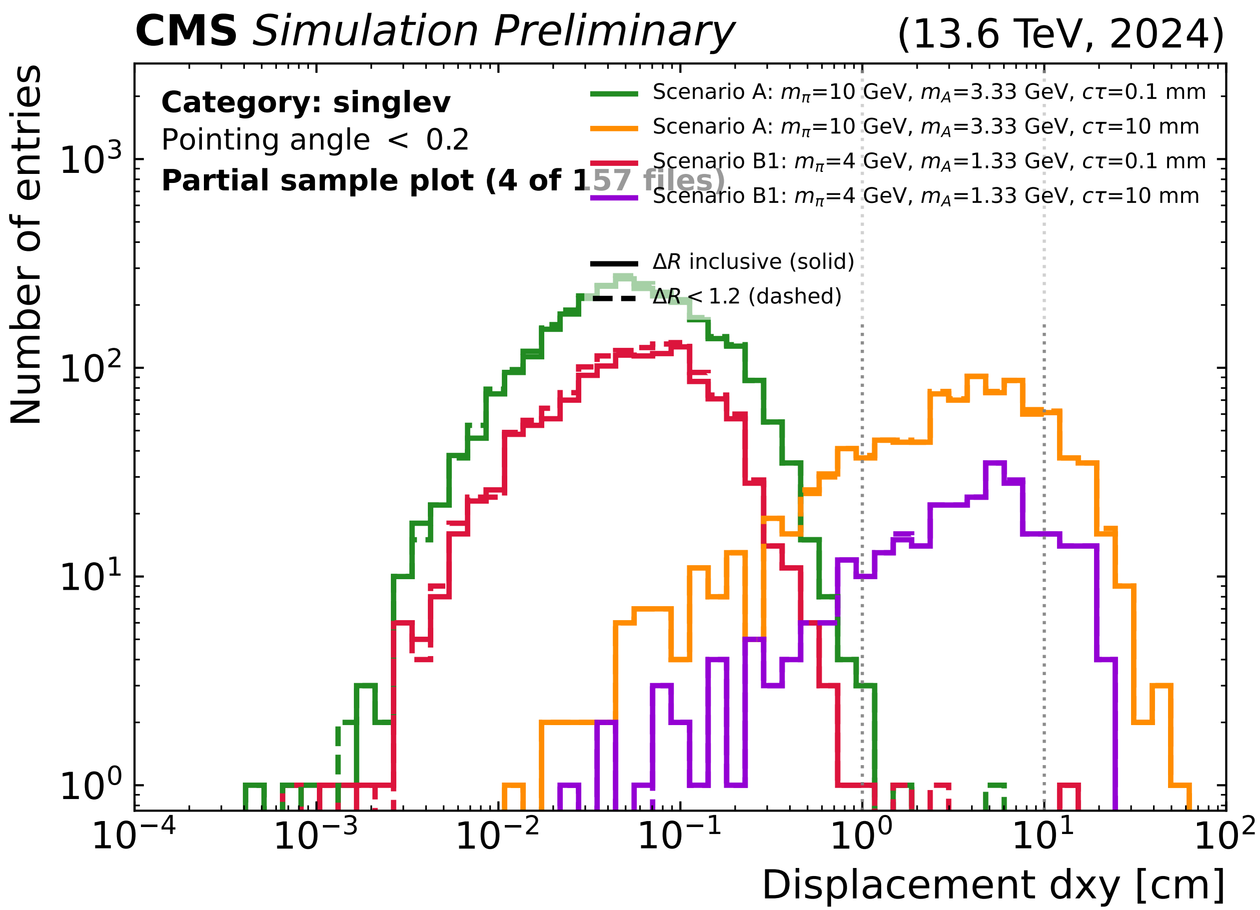
singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed);   dotted: dxy category edges (1, 10 cm)

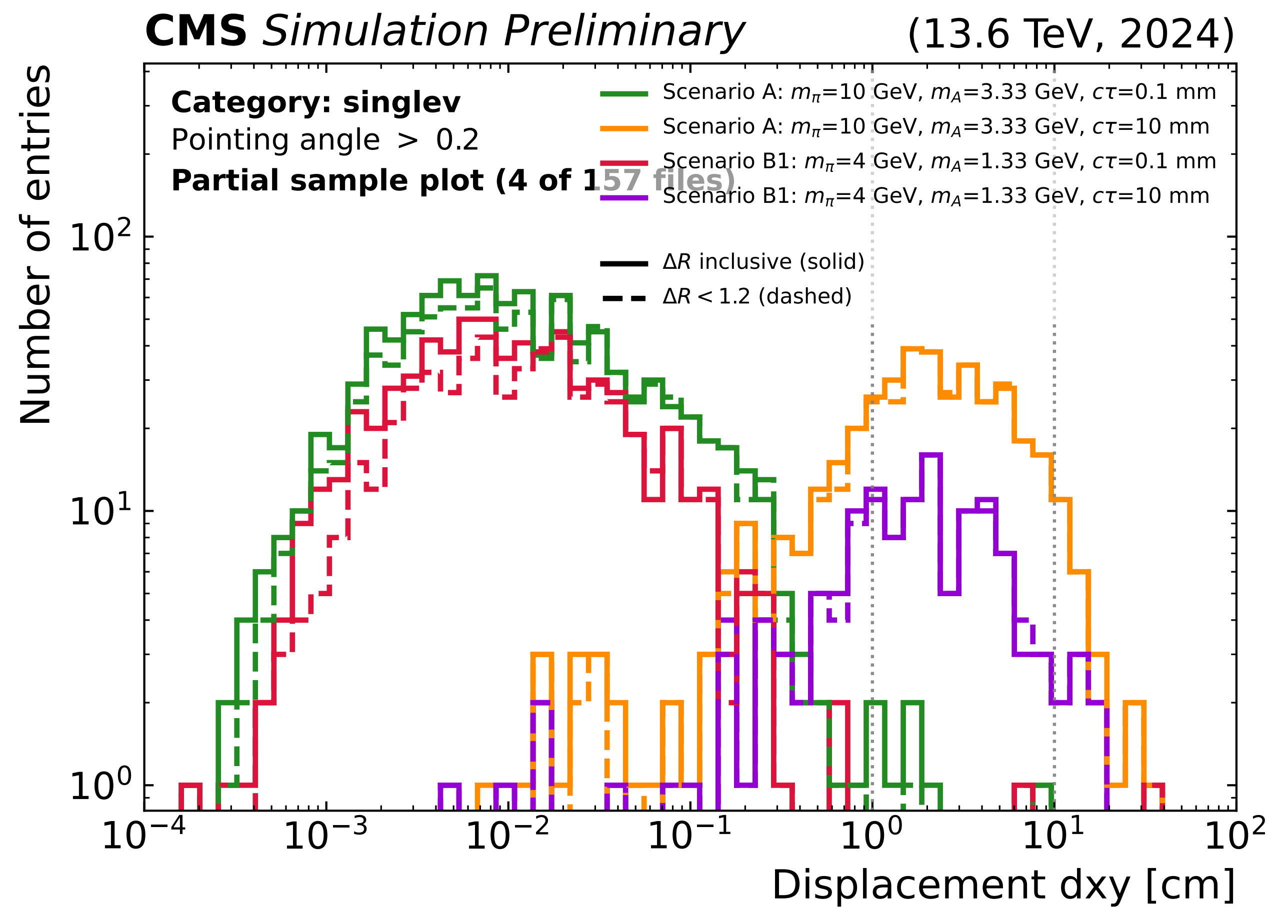


singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)



singlev: displacement of the binning vertex (muonSV_dxy at min_chi2_index), split by pointing angle
ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)



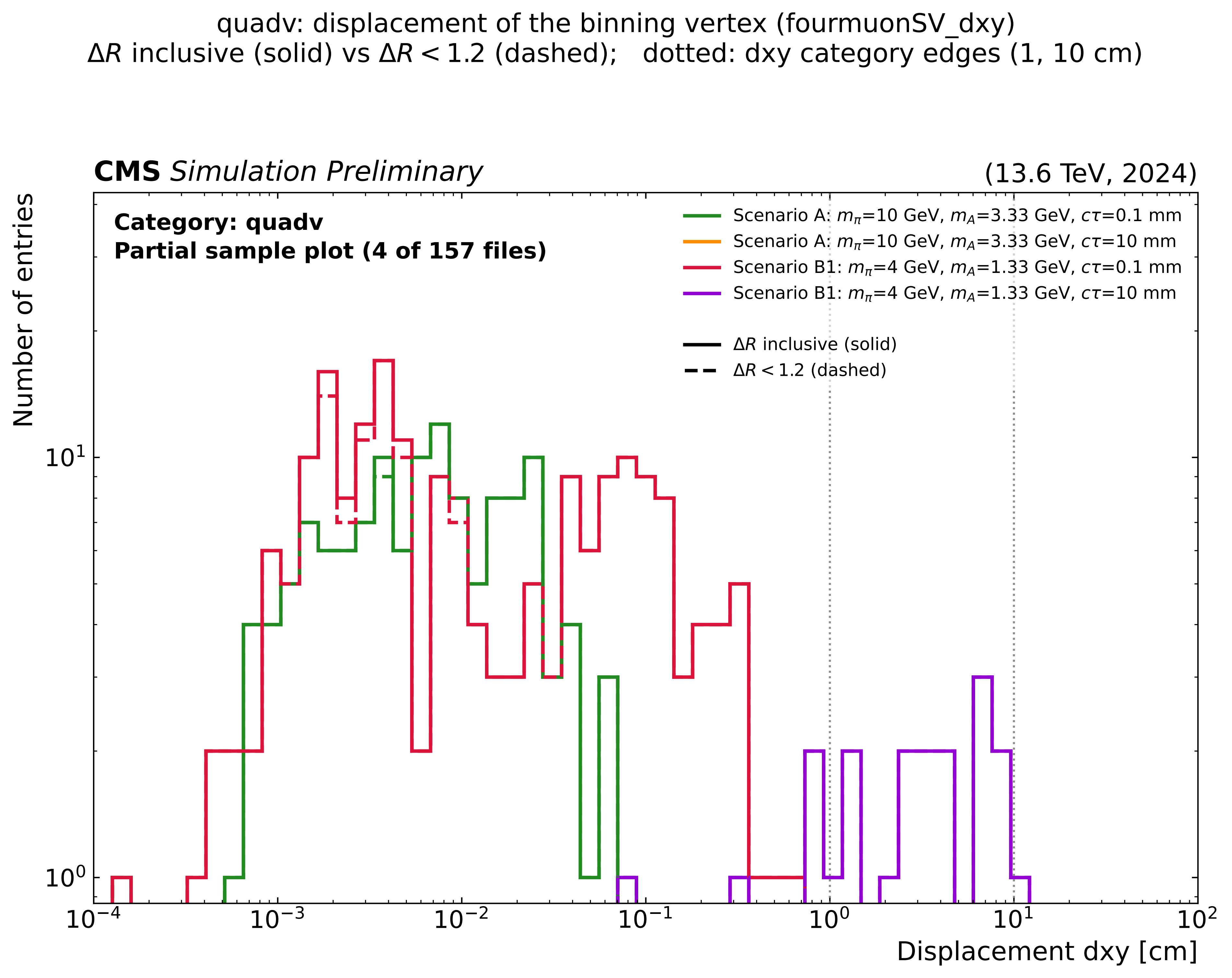


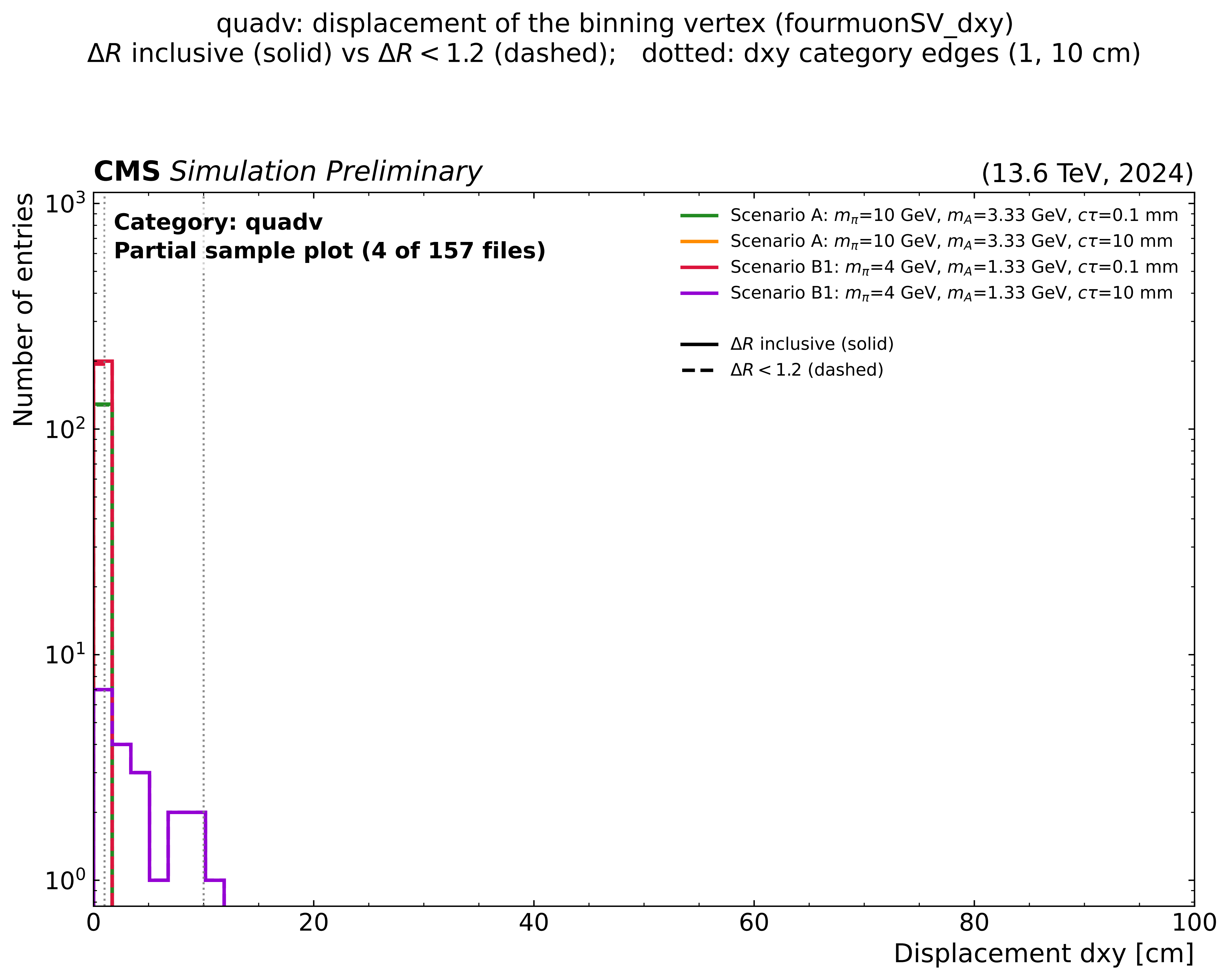
cell: N(ΔR inclusive) | N($\Delta R < 1.2$)

	dxy < 1	1--10	dxy > 10	Total
scA $c\tau$=0.1	2,796 | 2,883	2 | 3	0 | 0	2,798 | 2,886
scA $c\tau$=10	212 | 212	628 | 637	155 | 157	995 | 1,006
scB1 $c\tau$=0.1	1,260 | 1,350	0 | 2	1 | 1	1,261 | 1,353
scB1 $c\tau$=10	53 | 52	200 | 202	45 | 45	298 | 299

cell: N(ΔR inclusive) | N($\Delta R < 1.2$)

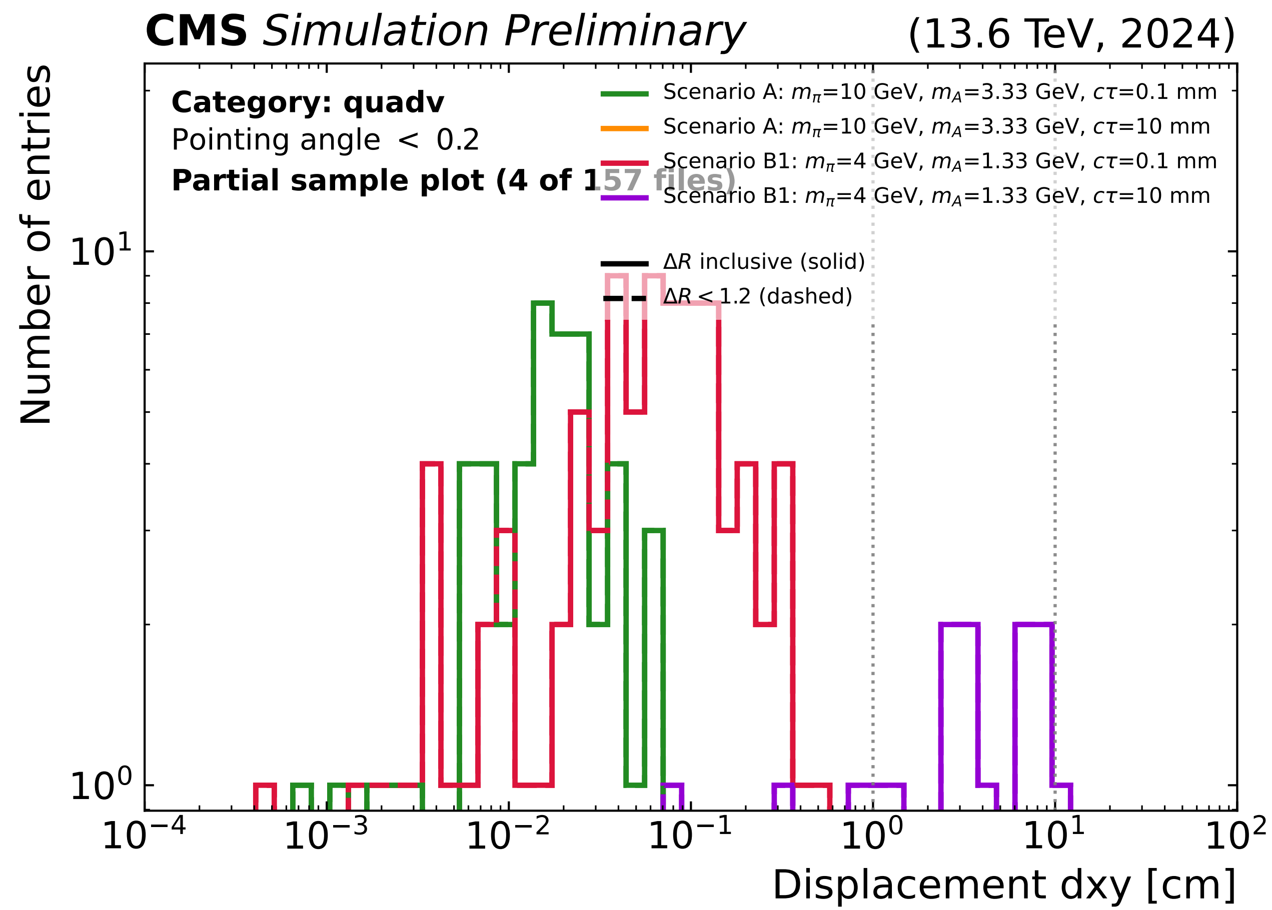
	dxy < 1	1--10	dxy > 10	Total
scA $c\tau$=0.1	1,010 | 901	6 | 2	0 | 0	1,016 | 903
scA $c\tau$=10	115 | 102	269 | 265	24 | 23	408 | 390
scB1 $c\tau$=0.1	659 | 547	1 | 1	1 | 1	661 | 549
scB1 $c\tau$=10	44 | 41	83 | 82	7 | 7	134 | 130

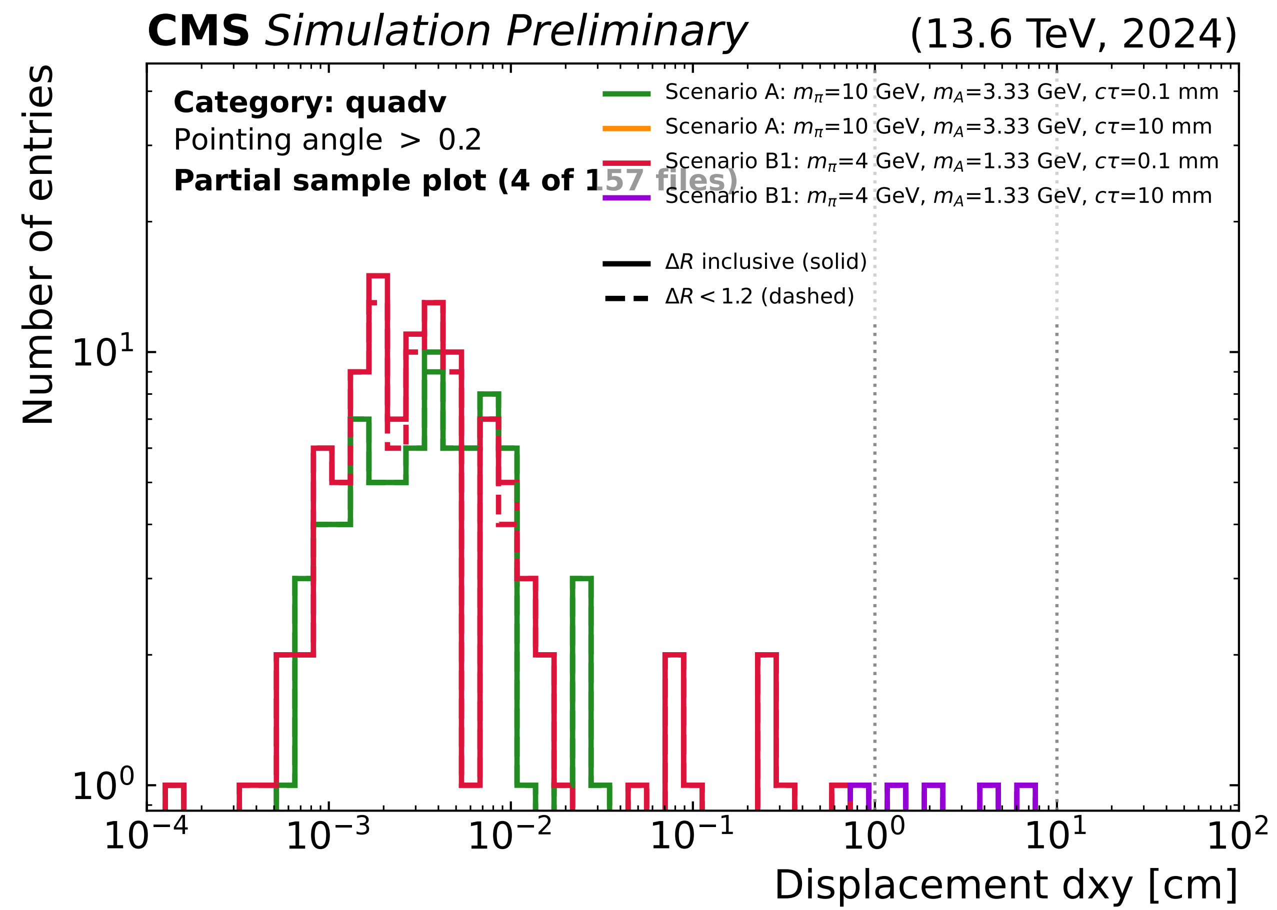

quadv: displacement of the binning vertex (fourmuonSV_dxy)
ΔR inclusive (solid) vs ΔR < 1.2 (dashed); dotted: dxy category edges (1, 10 cm)
CMS Simulation Preliminary
(13.6 TeV, 2024)
Category: quadv
Partial sample plot (4 of 157 files)
Scenario A: mπ=10 GeV, mA=3.33 GeV, cτ=0.1 mm
Scenario A: mπ=10 GeV, mA=3.33 GeV, cτ=10 mm
Scenario B1: mπ=4 GeV, mA=1.33 GeV, cτ=0.1 mm
Scenario B1: mπ=4 GeV, mA=1.33 GeV, cτ=10 mm
ΔR inclusive (solid)
ΔR < 1.2 (dashed)
Number of entries
Displacement dxy [cm]



quadv: displacement of the binning vertex (fourmuonSV_dxy)
ΔR inclusive (solid) vs ΔR < 1.2 (dashed); dotted: dxy category edges (1, 10 cm)
CMS Simulation Preliminary
(13.6 TeV, 2024)
Category: quadv
Partial sample plot (4 of 157 files)
Scenario A: m_π=10 GeV, m_A=3.33 GeV, $c\tau$=0.1 mm
Scenario A: m_π=10 GeV, m_A=3.33 GeV, $c\tau$=10 mm
Scenario B1: m_π=4 GeV, m_A=1.33 GeV, $c\tau$=0.1 mm
Scenario B1: m_π=4 GeV, m_A=1.33 GeV, $c\tau$=10 mm
ΔR inclusive (solid)
ΔR < 1.2 (dashed)
Number of entries
10^3
10^2
10^1
10^0
0
20
40
60
80
100
Displacement dxy [cm]

quadv: displacement of the binning vertex (fourmuonSV_dxy), split by pointing angle
ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)





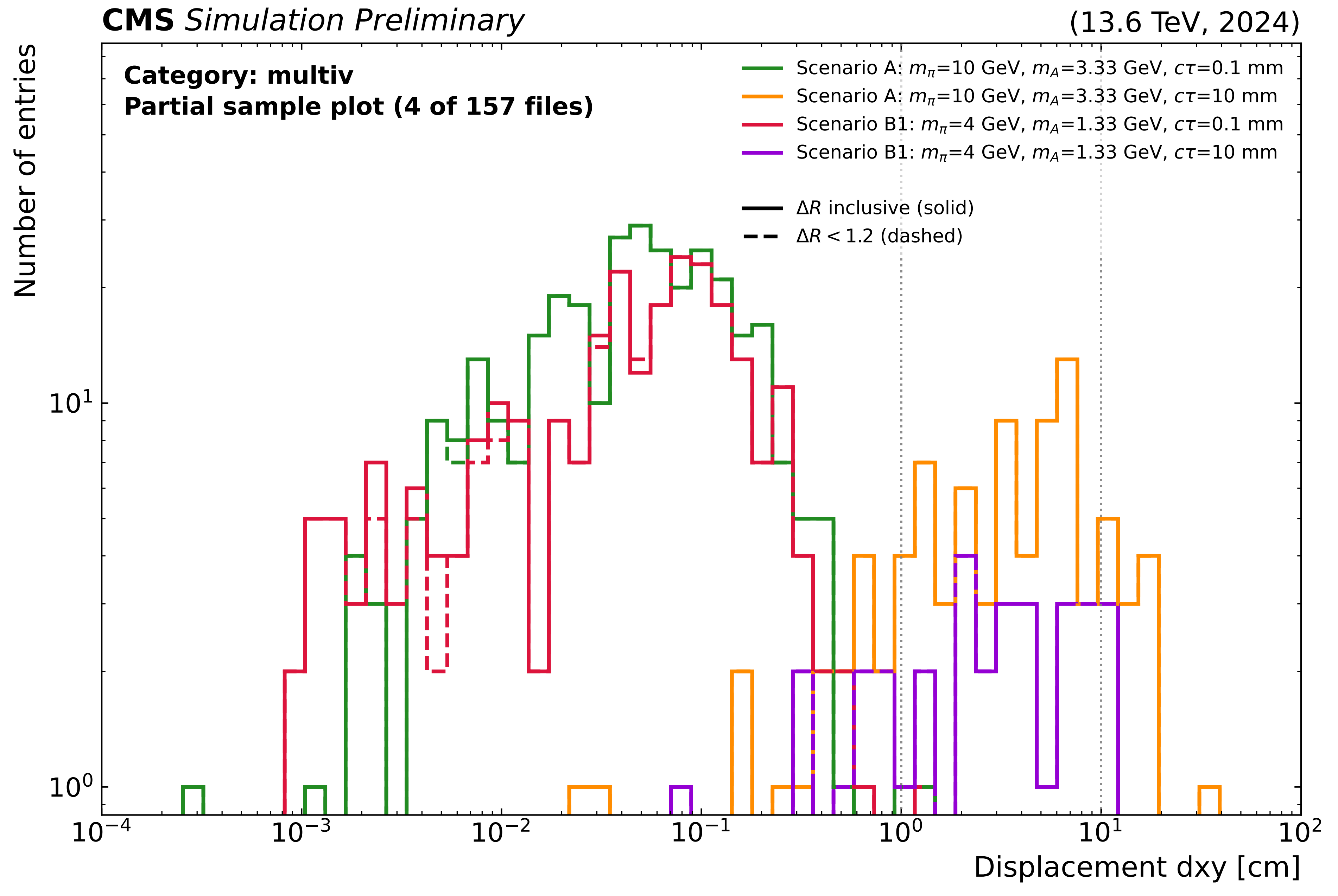
cell: N(ΔR inclusive) | N($\Delta R < 1.2$)

<table>
<tr><th></th><th>dxy < 1</th><th>1--10</th><th>dxy > 10</th><th>Total</th></tr>
<tr><td>scA cτ=0.1</td><td>51 | 51</td><td>0 | 0</td><td>0 | 0</td><td>51 | 51</td></tr>
<tr><td>scA cτ=10</td><td>0 | 0</td><td>0 | 0</td><td>0 | 0</td><td>0 | 0</td></tr>
<tr><td>scB1 cτ=0.1</td><td>90 | 90</td><td>0 | 0</td><td>0 | 0</td><td>90 | 90</td></tr>
<tr><td>scB1 cτ=10</td><td>3 | 3</td><td>11 | 11</td><td>1 | 1</td><td>15 | 15</td></tr>
</table>

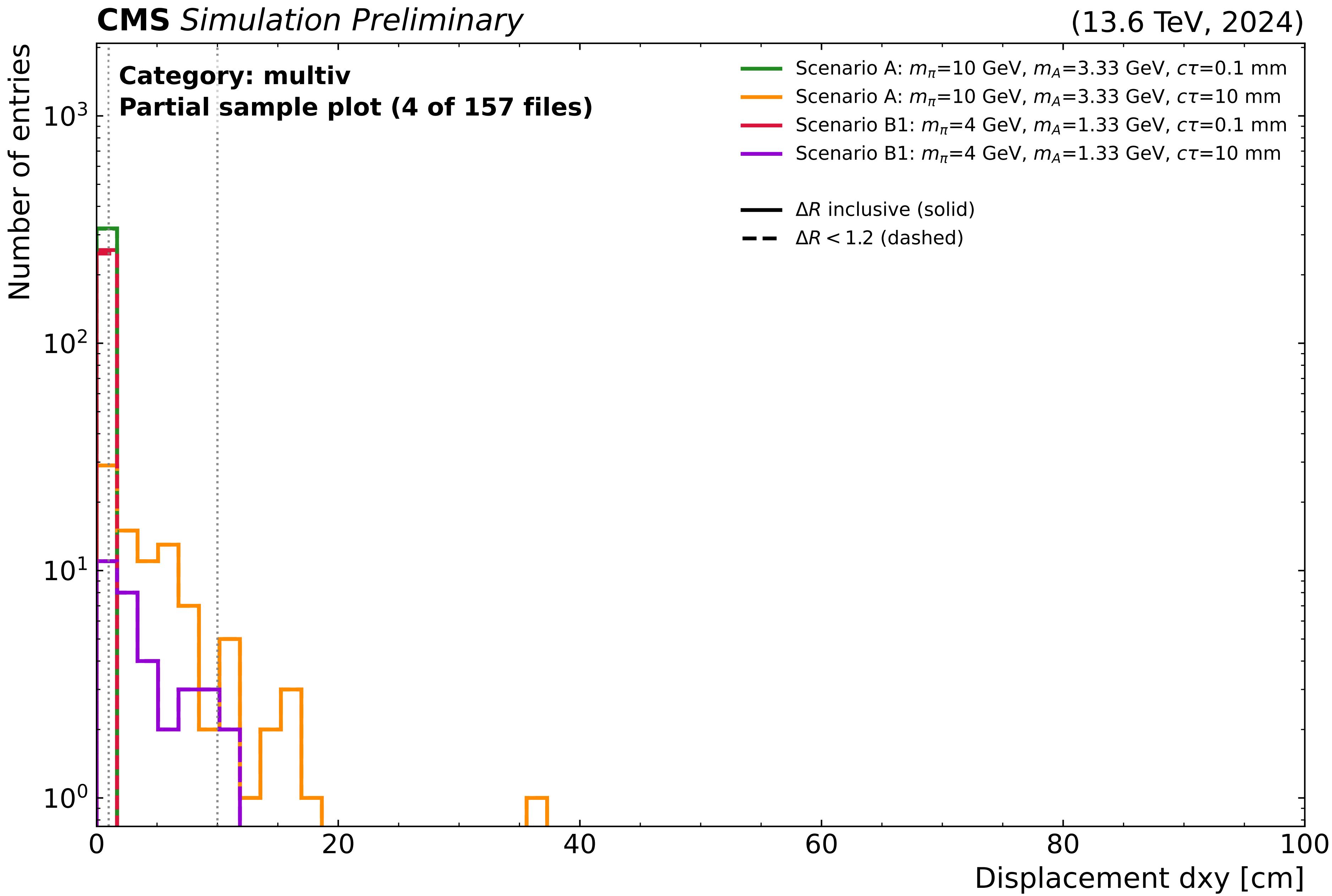
cell: N(ΔR inclusive) | N($\Delta R < 1.2$)

<table>
<tr><th></th><th>dxy < 1</th><th>1--10</th><th>dxy > 10</th><th>Total</th></tr>
<tr><td>scA cτ=0.1</td><td>78 | 77</td><td>0 | 0</td><td>0 | 0</td><td>78 | 77</td></tr>
<tr><td>scA cτ=10</td><td>0 | 0</td><td>0 | 0</td><td>0 | 0</td><td>0 | 0</td></tr>
<tr><td>scB1 cτ=0.1</td><td>110 | 104</td><td>0 | 0</td><td>0 | 0</td><td>110 | 104</td></tr>
<tr><td>scB1 cτ=10</td><td>1 | 1</td><td>4 | 4</td><td>0 | 0</td><td>5 | 5</td></tr>
</table>

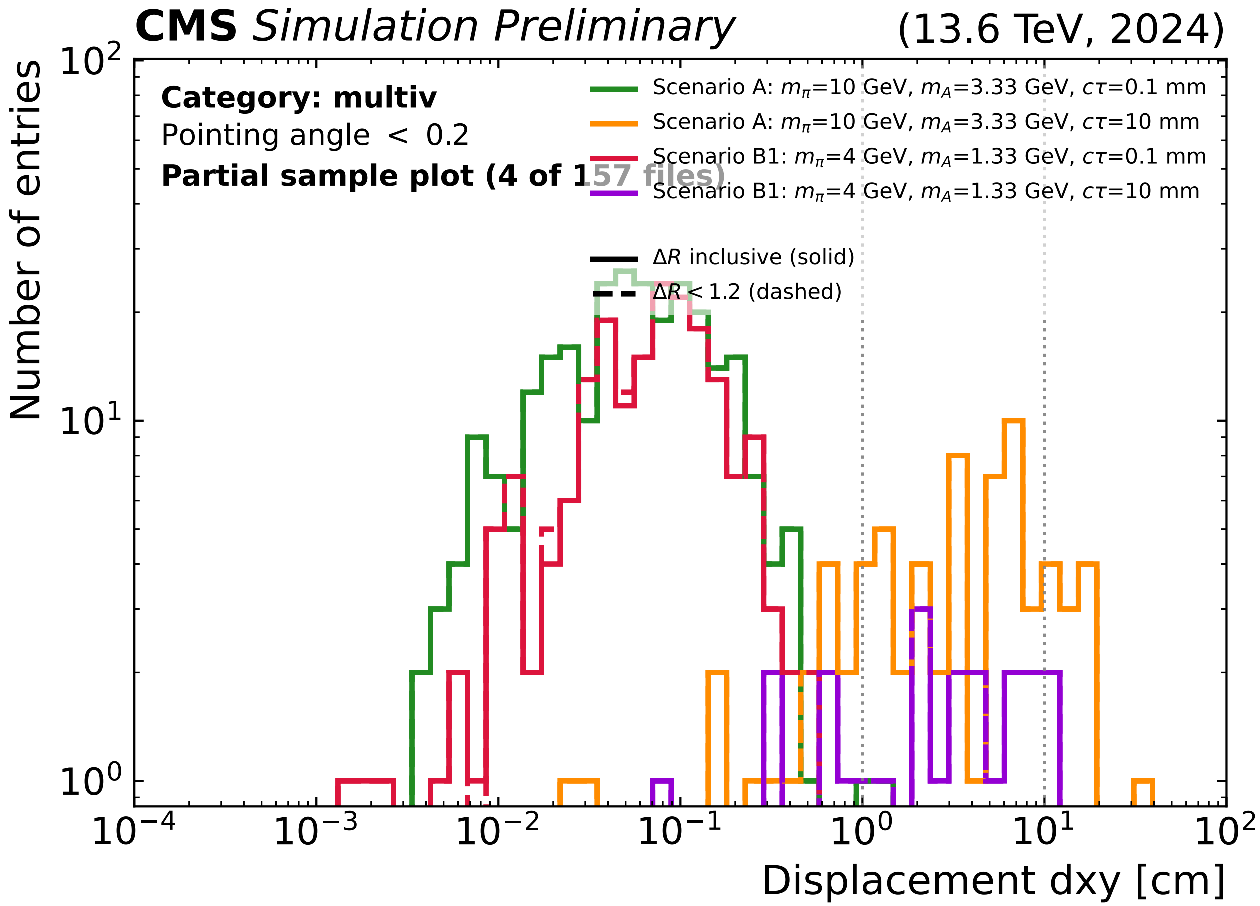
multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)

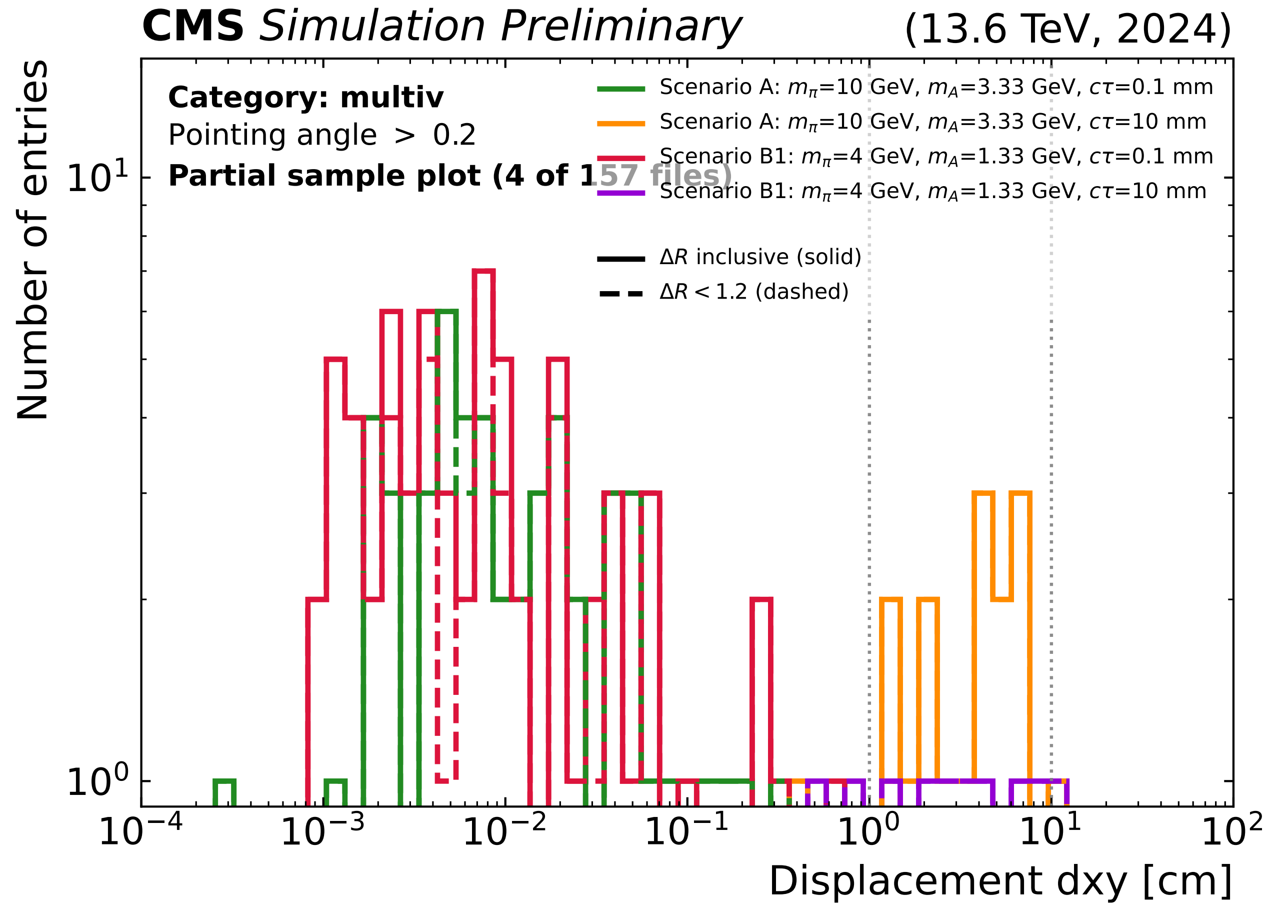


multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index)
ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)



multiv: displacement of the binning vertex (muonSV_dxy at min_chi2_index), split by pointing angle
ΔR inclusive (solid) vs $\Delta R < 1.2$ (dashed); dotted: dxy category edges (1, 10 cm)





cell: N(ΔR inclusive) | N($\Delta R < 1.2$)

	dxy < 1	1--10	dxy > 10	Total
scA cτ=0.1	266 | 266	2 | 2	0 | 0	268 | 268
scA cτ=10	17 | 17	44 | 44	12 | 12	73 | 73
scB1 cτ=0.1	189 | 190	0 | 0	0 | 0	189 | 190
scB1 cτ=10	6 | 6	15 | 15	2 | 2	23 | 23

cell: N(ΔR inclusive) | N($\Delta R < 1.2$)

	dxy < 1	1--10	dxy > 10	Total
scA cτ=0.1	52 | 51	0 | 0	0 | 0	52 | 51
scA cτ=10	1 | 1	15 | 15	1 | 1	17 | 17
scB1 cτ=0.1	67 | 58	1 | 0	0 | 0	68 | 58
scB1 cτ=10	2 | 2	7 | 7	1 | 1	10 | 10